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UNIVERSITY

ONLINE

## B. Tech – II

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Department of Science and  
Humanities



## **CLASS-4**

# **General property of LAPLACE TRANSFORM**



# Laplace transformation of derivatives/ multiplication by $s$

*Differentiation.*

Let  $\mathcal{L}\{f(t)\} = F(s)$ . Then

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0).$$

For the proof, consider

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t)e^{-st} dt$$

and use the integration by parts

$$\begin{aligned}
 \mathcal{L} \{ f'(t) \} &= e^{-st} f(t) \Big|_0^\infty + s \int_0^\infty f(t) e^{-st} dt \\
 &= -f(0) + s \mathcal{L} \{ f \} \\
 &= sF(s) - f(0).
 \end{aligned}$$

We can generalize this property to differentiation of any order.

For instance, to find  $\mathcal{L} \{ f''(t) \}$

just consider  $f''(t)$  as the derivative of  $f'(t)$  :

$f'(t)$  for which we already found the Laplace transform

Hence, according to the property,

$$\begin{aligned}
 \mathcal{L} \{ f''(t) \} &= s(sF(s) - f(0)) - f'(0) \\
 &= s^2 F(s) - sf(0) - f'(0).
 \end{aligned}$$

$$L\{f'''(t)\} = s^3 F(s) - s^2 f(0) - sf'(0) - f''(0)$$

And so the pattern continues.

# Laplace transformation of a integral

*If  $F(s) = \mathcal{L}(f(t))$  then*

$$\mathcal{L}\left[\int_0^t f(\tau) d\tau\right] = \frac{F(s)}{s}.$$

*Proof.* By definition, 
$$\mathcal{L}\left(\int_0^t f(\tau) d\tau\right) = \int_0^\infty e^{-st} \left(\int_0^t f(\tau) d\tau\right) dt$$
$$= \int_0^\infty \int_0^t e^{-st} f(\tau) d\tau dt.$$

order of the integrations can be changed and therefore

$$\int_0^\infty \int_0^t e^{-st} f(\tau) d\tau dt = \int_0^\infty \int_\tau^\infty e^{-st} f(\tau) dt d\tau.$$

$$\begin{aligned}
\mathcal{L}\left(\int_0^t f(\tau) d\tau\right) &= \int_0^\infty \int_0^t e^{-st} f(\tau) d\tau dt \\
&= \int_0^\infty \int_\tau^\infty e^{-st} f(\tau) dt d\tau = \int_0^\infty \int_\tau^\infty e^{-s(t-\tau)-s\tau} f(\tau) dt d\tau \\
&= \int_0^\infty e^{-s\tau} f(\tau) d\tau \left( \int_\tau^\infty e^{-s(t-\tau)} dt \right) \\
&= \int_0^\infty e^{-s\tau} f(\tau) d\tau \left( \int_0^\infty e^{-sz} dz \right) = F(s) \frac{1}{s}.
\end{aligned}$$

We can generalize this property :

$$L\left[\int \int \int \int \dots \int f(t) dt^n\right] = \frac{1}{s^n} L\{f(t)\}.$$

# Example

1. Find  $\mathcal{L}(\int_0^t \sin(az)dz)$ .

**Solution:** We know  $\mathcal{L}(\sin(at)) = \frac{a}{s^2 + a^2}$ . Hence

$$\mathcal{L}(\int_0^t \sin(az)dz) = \frac{1}{s} \cdot \frac{a}{(s^2 + a^2)}$$

### Division by $t$

If  $\mathcal{L}\{f(t)\} = F(s)$ , then,

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du$$

provided  $\lim_{t \rightarrow 0} \left[ \frac{f(t)}{t} \right]$  exists.

### Proof of Division by $t$

Let

$$g(t) = \frac{f(t)}{t}, \text{ then}$$

$$f(t) = t g(t)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{t g(t)\}$$

From [Multiplication by Power of  \$t\$](#)

$$\mathcal{L}\{f(t)\} = (-1)^1 \frac{d}{ds} \mathcal{L}\{g(t)\}$$

$$F(s) = -\frac{d}{ds} \mathcal{L}\{g(t)\}$$

$$F(s) ds = -d[\mathcal{L}\{g(t)\}]$$

$$d[\mathcal{L}\{g(t)\}] = -F(s) ds$$

Integrate both sides

$$\int d[\mathcal{L}\{g(t)\}] = - \int F(s) ds$$

$$\mathcal{L}\{g(t)\} = - \int_\infty^s F(u) du$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du$$





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## THANK YOU

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